# MILITARY SPECIFICATION

## RELAY ASSEMBLY, AIRPORT SERIES LIGHTING, 6.6 AMPERES

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

## 1. SCOPE

1.1 Scope.- This specification covers one type of relay assembly for use in airfield lighting circuits.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

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| MIL-C-5136 | Cable, Power, Electrical, Polychlorophrene Sheathed, Buna Compound Insulated |
| MIL-C-7192 | Connectors, Cable, Plug and Receptacle, Molded, and Connection Pins |
| MIL-P-15024 | Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment |
| MIL-E-17555 | Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment except Rotating Electrical Equipment |
| MIL-D-70327 | Drawings, Engineering and Associated Lists |

STANDARDS

Military

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| MIL-STD-108 | Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment |
| MIL-STD-129 | Marking for Shipment and Storage |
| MS33586 | Metals, Definition of Dissimilar |

(When requesting applicable documents refer to both title and number. Copies of this specification and applicable specifications and standards may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention: Code CDS.)

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## 3. REQUIREMENTS

3.1 Preproduction.- The relay assembly furnished under this specification shall be a product which has been tested and passed the preproduction inspection specified herein.

3.2 Materials.- Materials shall conform to applicable specifications and standards as specified herein. Materials which are not covered by applicable specifications or standards or which are not specifically described herein shall be of the best commercial quality and suitable for the purpose intended. The material shall be free from all defects and imperfections that may affect the quality and reliability of the finished product.

3.2.1 Protective treatment.- Materials used in the construction of the relay assembly that are subject to deterioration when exposed to climatic and environmental conditions during service usage, shall be suitably protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.2.2 Standard parts.- Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2.3 Metals.- All metals used in the construction of the relay assembly shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion. The use of dissimilar metals shall be avoided whenever practicable or used in accordance with Standard MS33586. The use of magnesium or any alloy thereof is prohibited.

3.2.4 All gaskets and other materials utilized in fabrication shall be of a type which will not emit smoke or vapor during rated operation of the relay assembly.

3.3 Design and construction.- The relay shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Construction shall be such that adjustments and repairs can be easily made by personnel of operating units and overhaul bases. No special tools shall be required for installation, adjustment or repair of the relay assembly.

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3.4 Component parts.- The relay assembly shall consist of the following major components:

A. Enclosure
B. Grounding lug
C. Relay
D. Leads

3.4.1 Details of components.-

3.4.1.1 Enclosure.- The enclosure shall be of cast metal, suitably protected, and shall be provided with a bracket for wall mounting. The enclosure shall include a removable cover and a gasket. The cover shall be fastened to the enclosure with not less than four screws; hinges are not required. The enclosure shall have two watertight bushings through which pass the leads to the relay. The enclosure shall be waterproof in accordance with Standard MIL-STD-108.

3.4.1.2 Grounding lug.- A lug, suitable for the attachment of a number 8 AWG grounding wire, shall be provided on the outside of the enclosure.

3.4.1.3 Relay.- The relay shall be a single-pole, single-throw, normally-open, heavy-duty, industrial control-type relay with a current sensing coil. The relay coil shall be operated at 6.6 amperes, 60 cycles, continuous duty. The voltage drop across the relay coil shall not exceed 4.0 volts at rated current and frequency. The relay shall pull in at 6.0 amperes and shall not drop-out at currents of more than 5.0 amperes. The rupture rating of the contacts shall be 250 volts, 45 amperes make, 45 amperes break at 60 cycles, for resistive loads. Moving parts of the relay shall be removable from the front. The moving contact shall be spring loaded in the closed position to provide positive contact and wiping action. The stationary contact shall be rigidly mounted. The relay terminals for external connection shall be screw type. The center-to-center spacing between adjacent terminals shall be at least 0.75 inch. The minimum contact air gap in the open position shall be 0.075 inch.

3.4.1.4 Leads.- Each relay shall be supplied with two 2-conductor leads. The leads shall pass through the watertight bushings of the enclosure. The conductors shall be securely fastened to the relay terminals. The leads shall be secured to the enclosure in such a manner that mechanical forces on the leads of the type encountered in service will not loosen the electrical connections or damage the water seals. All leads shall be constructed in accordance with Specification MIL-C-5136, or of similar high-quality commercial cable or cord having rubber or synthetic rubber insulation and a polychloroprene sheath. Each lead shall be 600 V, number 14 AWG flexible cord 72 ±3 inches long. Each conductor shall have not less than 37 strands. The



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conductors of one lead shall be terminated at the coil of the relay. The outer end of this lead shall terminate in a 600 V, 20 ampere, two-pole receptacle in accordance with figure 1. Part a, of Specification MIL-C-7192B. A watertight cap shall be placed upon the connector for protection during shipment and installation. The conductors of the other lead shall terminate at the contacts of the relay. The outer end of this lead shall be taped with insulation tape.

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.6 Identification of product.- A nameplate, permanently and legibly marked with the appropriate information shall be securely attached to the relay assembly. The information marked in the spaces provided shall be in accordance with Specification MIL-P-15024.

3.7 Performance.- When subjected to the tests specified in Section 4 under "Test methods," the relay assembly shall satisfy the requirements. Failure to do so shall be cause for rejection of the item.

3.8 Workmanship.- The relay, including all component parts, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alinement of parts and tightness of assembly screws, and bolts; etc.

3.8.1 Cleaning.- The relay assembly shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign matter removed during and after final assembly.

3.8.2 Preservation, packaging, packing and marking.- The inspector shall ascertain that the preservation, packaging, packing, and marking of relay assemblies conform to this specification.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.



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4.2 Classification of tests.- The inspection and testing of relay assemblies shall be classified as follows:

(a) Preproduction tests.- Preproduction tests are those tests accomplished on samples which are representative of the production of the item after award of the contract, to determine that the production meets the requirements of this specification.

(b) Acceptance tests: Acceptance tests are those tests performed on individual products or lots which have been submitted under contract to determine that the production meets the requirements of this specification.

4.3 Preproduction.-

4.3.1 *Preproduction tests.-* Preproduction tests shall consist of all the tests described under "Test Methods". Preproduction inspection may, at the option of the procuring activity, be supplemented with tests under actual service conditions.

4.3.2 Sampling instructions.- The contractor shall subject one complete relay assembly to the preproduction inspection. A complete test report referring to paragraph numbers of this specification, and indicating by substantial test data or evaluation that the relay satisfies all the applicable requirements of this specification, shall be furnished to the procuring activity prior to manufacture of any additional relays. The report shall be signed by the manufacturer or his authorized representative, or the director of the laboratory where the tests were conducted, as applicable. Reports shall be forwarded via the cognizant Government Inspector.

4.3.3 Preproduction inspection sample for the procuring activity.- When required by the procurement document, the contractor shall submit one complete relay assembly to the Government activity specified who will use it for the following:

(a) A review of the mechanical construction of the assembly.
(b) To perform any tests included in the specification after reviewing the contractor's test report.

4.4 Acceptance testing.- Acceptance testing shall consist of the Individual tests.

4.4.1 Individual tests.- Each relay assembly submitted for acceptance under contract shall be subjected to the following tests as described under "Test methods" of this specification. Relays not meeting the requirement of these tests may be corrected prior to submitting them for acceptance.

(a) Examination of product (4.6.1)
(b) Operational (4.6.10)
(c) Continuity (4.6.11)
(d) Dielectric (complete assembly) (4.6.12)



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4.5 Test conditions.- Tests shall be conducted at room temperature (approximately 20°C) unless otherwise specified.

4.6 Test methods.-

4.6.1 Examination of product.- Each relay assembly shall be inspected to determine compliance with the requirements of this specification with respect to design, materials, workmanship, and markings.

4.6.2 Dielectric (coil and contacts).- The relay shall withstand, without damage, dielectric tests in accordance with the following conditions:

(a) A potential of 2000 V d.c. shall be applied between the coil terminals and grounded metal parts for a period of one minute.

(b) With the relay contacts in the deenergized position, a potential equal to twice the rated contact voltage plus 2000 volts d.c. shall be applied between non-connected contact terminals and between all contact terminals and grounded-metal parts for a period of one minute. The same potential shall be applied between all contact terminals and grounded-metal parts for a period of one minute with the contacts in the energized position.

4.6.3 Pickup current.- The relay shall be tested to determine that the normally open contacts close when a current not in excess of 6 amperes is applied. An indicating device shall be used to determine that the contacts operate satisfactorily. The contacts shall not assume an intermediate position for any applied current.

4.6.4 Dropout current.- Specified operating current, 6.6 amperes, shall be applied to the holding-coil terminals. The current shall then be reduced. The relay shall release to the deenergized position within the specified limits of dropout current, 5 amperes. A device shall be used to determine that the contacts operate satisfactorily. The contacts shall not assume an intermediate position when the coil current is reduced from maximum to the dropout current.

4.6.5 Coil voltage drop.- The relay coil shall be subjected to the specified operating current, applied to the coil terminals for at least one hour. The voltage drop across the coil for this current shall be determined. The voltage drop shall not exceed 4.0 volts.

4.6.6 Contact drop.- The voltage drop across relay contacts, as measured at the relay contact terminals, shall be determined at rated-resistive load through the contacts and minimum coil current. The drop shall not exceed 0.1 volt as determined from an average of 10 consecutive measurements.



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4.6.7 Rupture.- The relay contacts shall be made to make and break 45 amperes at rated voltage for 50 operations. The on-period shall be 0.2 ±0.005 second, and the off-period shall be 30.0 seconds. No failure shall occur as a result of this test. Following this test, the relay shall meet the test specified in 4.6.6.

4.6.8 Continuous operation.- The relay shall be subjected to continuous operation for 100 hours with rated operating current through the coil and with rated resistive load through the contacts. Immediately after the test, the relay shall be subjected to the test specified in 4.6.6 and shall meet the requirements specified therein.

4.6.9 Life test.- The relay shall be subjected to 20,000 operating cycles at rated Resistive Load and Voltage. There shall be no failure during this test. After this test the relay shall be subjected to the test specified in 4.6.6 and shall meet the requirements specified therein.

4.6.10 Operational.- Each relay shall be completely assembled, mounted in normal position, and operated, to determine that components parts fit properly, and that design and performance requirements are met.

4.6.11 Continuity.- Each relay assembly shall be subjected to continuity tests to determine that the wiring connections are correct and that the contacts make and break properly.

4.6.12 Dielectric (complete assembly).- Each relay assembly shall be subjected to a dielectric test conducted at 1,500 V a.c. or 2,000 V d.c. to determine that each terminal, and the wires connected thereto, is properly insulated from ground and from all other wires.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing of relay assemblies shall be in accordance with applicable levels of Specification MIL-E-17555, as specified in the procurement document (see 6.2).

5.2 Marking.- In addition to any special marking required by the procurement document, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

## 6. NOTES

6.1 Intended use.- The relay assembly is intended for use in airfield lighting circuits where a light in one constant-current series circuit is automatically cutout when a second series circuit is energized. The relay coil is supplied with power from the second series circuit through a 30/45 watt, 6.6/6.6 ampere isolating transformer. The normally open contacts of the relay are connected across a light which is powered from a similar isolating transformer having its primary connected in the first series circuit.



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6.2 Ordering data.- Procurement documents should specify the following.

(a) Title, number and date of this specification.
(b) Quantity of relay assemblies required.
(c) Selection of applicable levels of preservation, packaging and packing (see 5.1.1).

6.3 Provisions for preproduction tests.- Manufacture of any further relay assemblies on the particular contract shall be suspended until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these relay assemblies from contract to contract submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.3.1 It shall be understood that the product supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship, and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the Procuring Activity. Evidence of unauthorized change shall consitute cause for rejection.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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